PATHOCLASS-BRCA



Free-Text Reports VERBOSE!

"Microscopic observation of the pathology slide reveals minimal formation of glandular structures. The nuclei exhibit moderate variability in size and shape, and a low rate of cell division is observed. Lymphatic invasion is not present. Based on these observations, the diagnosis is invasive ductal carcinoma, Nottingham Histologic Grade 2."

Structured Report ICCR ALIGNED

Histological Type	Invasive Ductal Carcinoma
Histological Grade	Grade 2
Nuclear Pleomorphism	Moderate
Mitotic Activity	Low
Tubular Formation	Low
LVI	Absent

BLEU-4 0.87/1.0

Correct 3/6

Pathological Feature-based Evaluation

Traditional Generation Model

Caption-based Evaluation NOT CLINICALLY MEANINGFUL!

"Microscopic observation of the pathology slide reveals high formation of glandular structures. The nuclei exhibit low variability in size and shape, and a low rate of cell division is observed. Lymphatic invasion is not present. Based on these observations, the diagnosis is invasive ductal carcinoma, Nottingham Histologic Grade 1."

Our Evaluation CLINICALLY MEANINGFUL

Histological Type	Invasive Ductal Carcinoma	✓
Histological Grade	Grade 1	✗
Nuclear Pleomorphism	Low	✗
Mitotic Activity	Low	✓
Tubular Formation	High	✗
LVI	Absent	✓

Correct 5/6

Classification Framework

Classification Model

- Direct prediction of pathological features (No free-text report generated)
- Increased Clinical Correctness
- ICCR GUIDELINE-ALIGNED

Predicted Pathological Features

Histological Type	Invasive Ductal Carcinoma	✓
Histological Grade	Grade 2	✓
Nuclear Pleomorphism	Moderate	✓
Mitotic Activity	Low	✓
Tubular Formation	Moderate	✗
LVI	Absent	✓